

Net731: Week 4 Bongani Thengeni

20240038

Task 01: Planning Your Identity Structure

Before opening the portal, we define our standards. Using a consistent naming convention is vital in enterprise environments to ensure uniqueness, simplify automation scripts, and to make it easy for admins to identify users.

Identity Design Table

Department	User Example (Display Name)	Username (UPN)	Group Name	Assigned Role
Development	Alex Mokoena	alex.mokoena@bthdomain.onmicrosoft.com	project-dev	Contributor
Operations	Sam Taylor	sam.taylor@bthdomain.onmicrosoft.com	project-op	Reader
Development	Jordan Nkosi	jordan.nkosi@bthdomain.onmicrosoft.com	proj-dev	Contributor
Finance	Morgan Smit	morgan.smit@bthdomain.onmicrosoft.com	proj-finance	Billing Reader

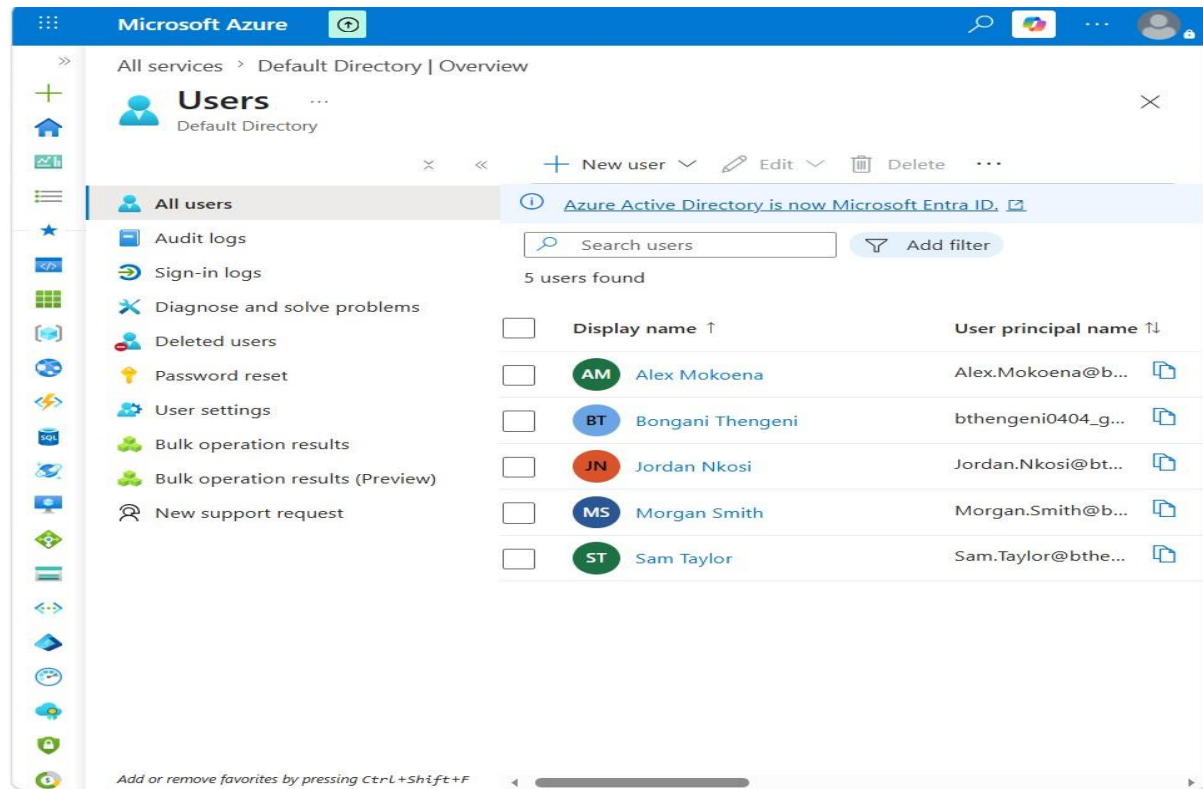
We use **Security Groups**. Unlike Microsoft 365 groups (which include shared mailboxes and calendars), Security groups are designed specifically for granting access to resources like Azure VNets or Subscriptions via RBAC.

Task 02: Create Users and Groups in Entra ID

2a: Provisioning Users

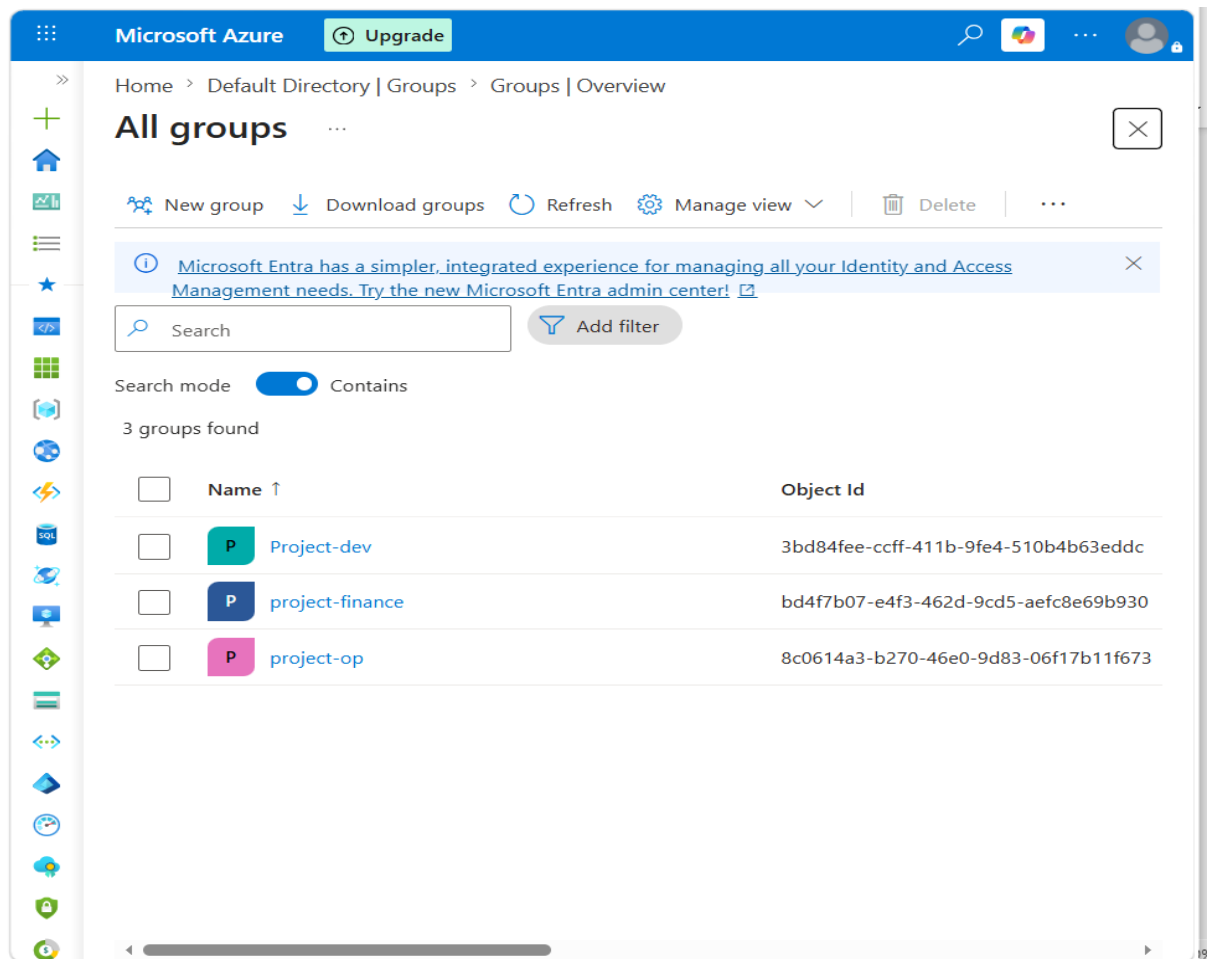
1. Log in to the **Azure Portal**.
2. Search for and select **Microsoft Entra ID**.
3. Under the **Manage** menu on the left, select **Users > New user > Create new user**.
4. Fill in the details based on your plan:

- **User principal name:** firstname and lastname
 - **Display name:** Full Name
 - **Job title & Department:** Developer/ Assistant
5. **Important:** Select **Review + create** and copy the temporary password. Repeat this for all 6 users.



2b: Creating Groups

1. In Entra ID, select **Groups > New group**.
2. **Group type:** Security.
3. **Group name:** proj-dev (repeat for ops and finance).
4. **Membership type:** Assigned.
 - *Note:* Dynamic membership is better for large organizations because it automatically adds users based on reducing manual admin work.
5. Click **Members > Add members** and select the two users you created for that specific department.
6. Click **Create**.



Task 03: Assign Initial Roles (RBAC)

Assigning roles to groups is the "Golden Rule" of IAM. It helps to ensure that when a new employee joins, you simply add them to a group, and they automatically inherit the correct permissions specified by the group.

Step-by-Step Assignment

1. Navigate to the **Resource Group** or **Subscription** you are using for Project .
2. Select **Access control (IAM)** from the left sidebar.
3. Click **+ Add > Add role assignment**.
4. **Select the Role:** "Contributor" for the Dev group.
5. **Members:** Choose "User, group, or service principal" and click **+ Select members**.
6. Search for your group (project-dev) and select it.
7. Click **Review + assign**.

Microsoft Azure

Dashboard > Techbridge-RG | Access control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

Role

Contributor

Scope

/subscriptions/99a5f00b-d7d7-4b59-886f-83b5d137bc38/resourceGroups/Techbridge-RG

Members

Name	Object ID	Type
Alex Mokoena	57f104dd-8df2-465c-b4d1-3e2319ec4f3a	User

Description

No description

Review + assign

Previous

Next

Feedback

Microsoft Azure

Dashboard > Resource Manager | Resource groups > Techbridge-RG | Access control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

Role

Billing Reader

Scope

/subscriptions/99a5f00b-d7d7-4b59-886f-83b5d137bc38/resourceGroups/Techbridge-RG

Members

Name	Object ID	Type
Morgan Smith	2c959e6e-af3c-4487-8d8c-633303ea44b8	User

Description

No description

Review + assign

Previous

Next

Feedback

Microsoft Azure

Dashboard > Resource Manager | Resource groups > Techbridge-RG | Access control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

Role

Contributor

Scope

/subscriptions/99a5f00b-d7d7-4b59-886f-83b5d137bc38/resourceGroups/Techbridge-RG

Members

Name	Object ID	Type
Jordan Nkosi	f93504da-e488-406c-8863-be03d2a3dba3	User

Description

No description

Review + assign

Previous

Next

Feedback

Microsoft Azure

Dashboard > Resource Manager | Resource groups > Techbridge-RG | Access control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

Role

Reader

Scope

/subscriptions/99a5f00b-d7d7-4b59-886f-83b5d137bc38/resourceGroups/Techbridge-RG

Members

Name	Object ID	Type
Sam Taylor	93fc743e-ba2d-4d4b-b6bd-dca582354818	User

Description

No description

Review + assign

Previous

Next

Feedback

Scope Choice: For this project, assigning at the **Resource Group** level is usually best.

- **Broader Scope (Subscription):** Easier to manage but risks "over-privileging" (violating Least Privilege).
- **Narrower Scope (Resource):** Very secure but creates a massive administrative burden as the environment grows.